Fluid Dynamics and Bernoulli's Principle

Synopsis: Note 1 was fluid standing still. Everything there came from one idea – the fluid above you has weight. Now we set the fluid moving, and two new ideas take over, both of them conservation laws you already know, rewritten for something that flows.

Mass is conserved, so whatever goes into a pipe has to come out of it. Squeeze the pipe and the fluid has no choice but to speed up – that is the continuity principle.

Energy is conserved, so a fluid that speeds up has to pay for the extra kinetic energy out of something else. What it pays with is pressure. That is Bernoulli's equation, and it is the work–energy theorem with the terms divided through by volume.

Put the two together and you get the result that sounds wrong the first time you hear it: where a fluid moves faster, its pressure is lower. It is why a wing lifts, why a shower curtain gets sucked inward, and why the narrow part of a pipe is the part at risk of collapsing.

Performance Expectations: State the continuity principle and use $A_1v_1 = A_2v_2$. Explain where Bernoulli's equation comes from, term by term, and say what each term is the energy of. Use it to compare two points on a streamline. Explain why faster flow means lower pressure, and apply that to a wing.

For PHY 151 — with calculus: Get Bernoulli's equation by integrating Euler's equation along a streamline, see exactly which assumptions the integral needs, and recover Torricelli's result for a draining tank as a special case.

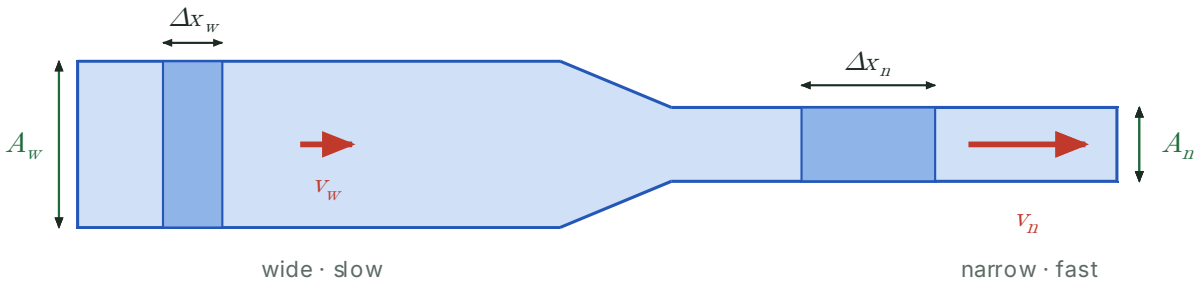
Before you read on – predict. A pipe carrying water narrows to half its width and then opens out again. In the narrow section, is the water pressure higher than in the wide section, lower, or the same? Commit to one here, with a reason, before you watch. You will come back to it at the end.

Higher / lower / the same — and why

Instructions and Questions

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1) Continuity – the fluid has to go somewhere.
   Water goes into one end of a pipe and comes out of the other. None of it is created on the way through and none of it disappears, so in any given second, the amount that enters has to equal the amount that leaves. That is mass conservation, and for a fluid of constant density it is volume conservation too.



the two shaded slugs hold the same volume — and they take the same time to pass

Mark A_w, A_n and both slugs on the sketch. The two shaded slugs hold the same volume – that is the whole content of the principle. Now write it down:

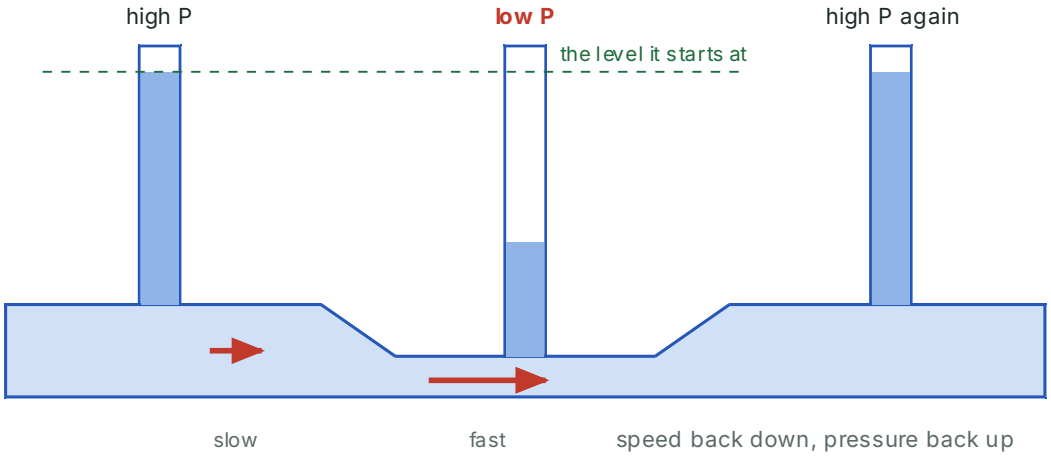
in a time Δt the wide slug travels Δx_w and the narrow slug travels Δx_n, and the volumes match:
_____ = _____
divide by Δt: _____ = _____
_____ = _____ because Δx/Δt **is** the speed

i) That product Av has a name and a unit. What is it called, and what is it measured in?
ii) Write the see-saw in words: if A goes up, what must v do, and why can it not do anything else?
iii) You put your thumb over the end of a hose and the water goes further. Which quantity did you change, and which one responded?

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2) Fast flow is low-pressure flow.

Here is where intuition goes wrong. Squeezing a pipe sounds like it should raise the pressure inside the squeeze – you are cramming the same water through a smaller hole. The opposite happens. The narrow section is the **low**-pressure section.



The standpipes read the pressure in the duct below them. Mark which column is the short one, and mark where the pressure **recovers**. Then settle these three:

Claim	**True or false, and why**
Squeezing the pipe raises the pressure in the squeeze	
The fluid speeds up because something pushed it harder	
Energy is lost at the throat	

Write the chain out, and name the conservation law that gives each arrow:

narrow → fast → low pressure

first arrow: _____

second arrow: _____

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3) Bernoulli's equation – energy conservation for a fluid.
Take the work–energy theorem you used for blocks and carts in Module 4, apply it to a lump of fluid, and divide every term by the lump's volume. Nothing new goes in; what comes out is Bernoulli's equation. The surrounding fluid pushes the lump along with a force PA over a distance – that is a work of P × volume.

$\frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 = -[\,(PE_f - PE_i) + $ _____ $\,]$ ← label this term: the work done by the _____ divide every term by V: _____ and m/V is _____ : _____ _____ = _____

Name the three terms, and say what each one is the energy of — and what they are all divided by:

Term	Its name	The energy of what
$\frac{1}{2}\rho v^2$		
ρgh		
P		

i) One of those three ran the whole of Note 1. Which one, and what happens to Bernoulli's equation if you switch the flow off ($v_i = v_f = 0$)?

A horizontal pipe narrows. The water moves at 2.0 m/s in the wide part, where the pressure is 150 kPa, and the narrow part has half the area. Find the pressure in the narrow part.	**Your working**
$\rho = 1000\ kg/m^3$	

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i) Which did you have to use first, continuity or Bernoulli? Say why that is the order for almost every problem of this kind.
ii) The drop came out small – a few kPa out of 150. Which term is responsible, and what would have to change to make the effect dramatic?

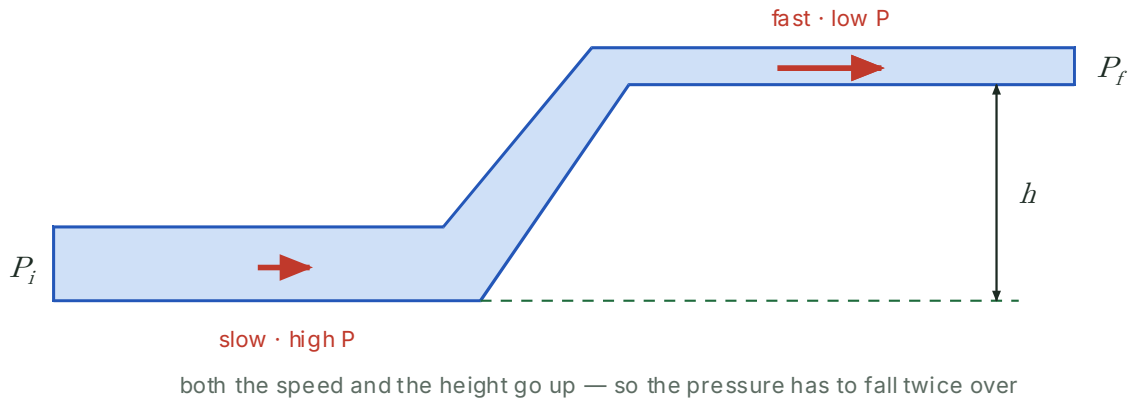
Bernoulli as an integral along a streamline.

For steady flow of an inviscid, incompressible fluid, Newton's second law applied to a fluid element along its own streamline is Euler's equation, $\rho v\, dv/ds = -dP/ds - \rho g\, dh/ds$, where s measures distance along the streamline. Multiply through by ds and integrate from one point to another: every term is an exact differential, so the integral collapses to $\frac{1}{2}\rho v^2 + P + \rho g h = \text{constant}$. Written this way the assumptions are visible rather than buried – steady, inviscid, incompressible, and along one streamline. Every failure of Bernoulli you will meet is one of those four being false.

i) A tank of water with a small hole a depth d below the surface. Apply Bernoulli from the surface to the hole, argue that both pressures are atmospheric and the surface speed is negligible, and obtain Torricelli's result $v = \sqrt{(2gd)}$.
ii) That is the same speed as an object dropped from height d. Say why, in terms of which Bernoulli terms survived.
iii) The tank drains, so d falls. Write dd/dt in terms of the hole area a, the tank area A and d, and say what shape of curve d(t) is.

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4) When the pipe also rises.
So far the pipe has been flat and the ρgh terms have cancelled. Let them not cancel. Here the pipe narrows **and** climbs at the same time, so the fluid gains kinetic energy and gains potential energy – and both bills land on the same account.



both the speed and the height go up — so the pressure has to fall twice over

Mark h and the datum on the sketch. Then take Bernoulli, put the kinetic terms on the left and everything else on the right, and read the signs:

$\frac{1}{2}\rho v_f^2 - \frac{1}{2}\rho v_i^2 = -[$ _____ + _____]

ΔKE = −[ΔPE + ΔP] all three per unit volume

ΔKE is _____ because _____

ΔPE is _____ because _____

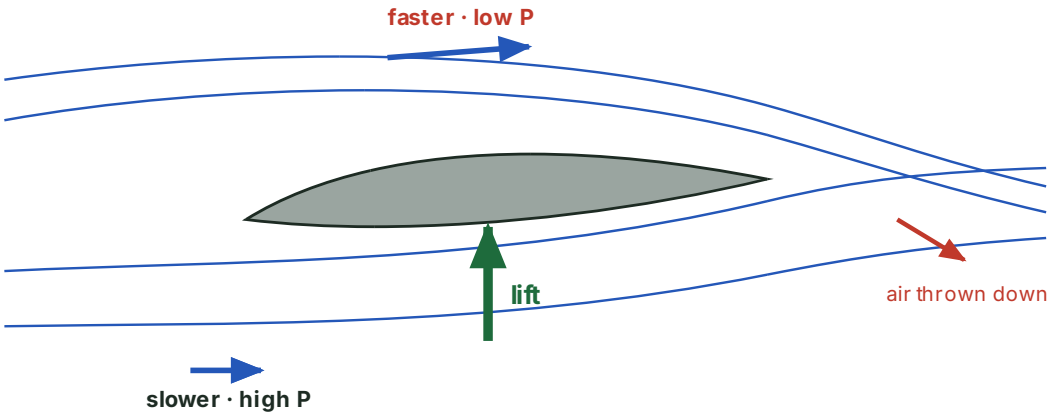
so ΔP must be _____ , and _____ enough to outweigh the climb

i) Compare with the same pipe kept flat. Is the pressure drop larger or smaller here, and why?
ii) Why does the top floor of a building have weaker water pressure than the ground floor? Answer in terms of the three terms above.

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5) The wing.

The air passing over the upper surface of a wing moves faster than the air passing underneath. By Bernoulli, faster air is at lower pressure. So the pressure underneath exceeds the pressure above, and that pressure difference, acting over the whole area of the wing, is lift.



Mark the pressures above and below, and mark the lift. Note that the streamlines over the top are drawn crowded – crowded streamlines are fast-moving fluid, which is the same continuity idea as the narrow pipe.

Why is the air over the top moving faster?	**Your answer**
"Because it has further to go and has to meet the air from underneath at the trailing edge." Is that right?	

i) Write one line for the Bernoulli account of lift and one line for the Newton account.
ii) Are those two competing explanations, or the same lift counted twice? Say which, in one sentence.
iii) A note to yourself: what is the thing about wings that you should **not** repeat?

Back to your prediction. Look at what you wrote at the top of this handout. Was it right? If you said "higher", write one sentence saying what the squeeze actually does.

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Check yourself

Four to do once you have filled the notes in. Work them here, then check them against the interactive version at the bottom of the note page.

1) Water flows at 3.0 m/s through a pipe of cross-sectional area 0.020 m^2. The pipe narrows to 0.0050 m^2. What is the speed in the narrow section, in m/s?

2) In that narrow section, compared with the wide section, the pressure is
   i) higher, because the same water is forced through a smaller opening
   ii) lower, because the water is moving faster there
   iii) the same, because the pipe is horizontal
   iv) impossible to say without the density

Why?

3) Which term of Bernoulli's equation is the one that carries over unchanged from the fluid statics of Note 1?
   i) $\frac{1}{2}\rho v^2$
   ii) $\rho g h$
   iii) P
   iv) none of them – statics is a separate result

Why?

4) A pipe narrows **and** rises. Taking $\Delta KE = -[\Delta PE + \Delta P]$, the pressure change ΔP is
   i) positive, and small
   ii) negative, and smaller in size than for the same pipe kept flat
   iii) negative, and larger in size than for the same pipe kept flat
   iv) zero, because the two gains cancel

Why?

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Your notes

Space for the sketches, the derivations you want to redo yourself, and anything from the video that did not fit above. Continuity and Bernoulli close Module 6; the method of Note 1 – weighing a sample twice – is what you carry into the Buoyancy Lab.

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